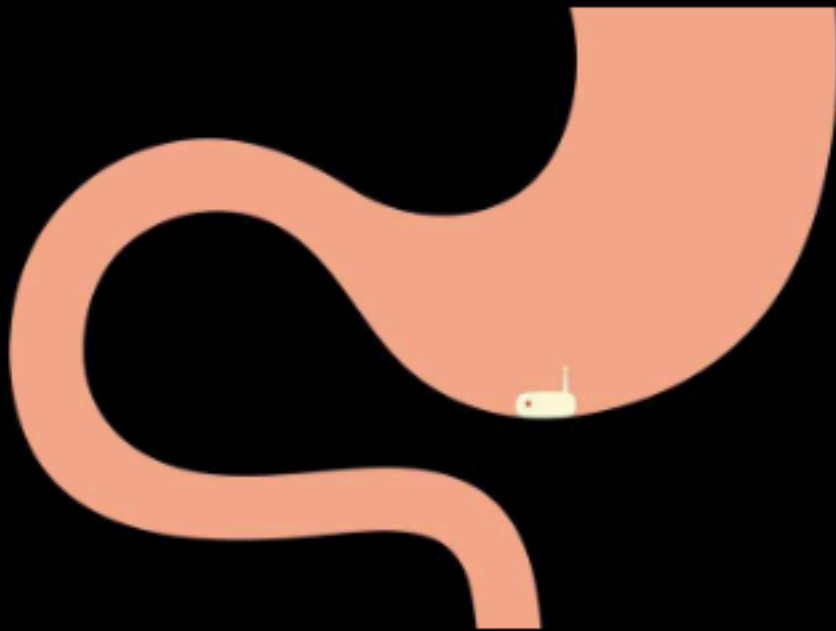


Science & technology | Edible electronics

Tomorrow's medical sensors might come served with dinner

Listening your gut could involve riboflavin batteries and toothpaste transistors

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Swallowing electronics is not usually recommended. But researchers in Belgium and the Netherlands have worked out how to make eating components, from a wireless transmitter and microchips to a battery and a suite of chemical sensors, not just safe but useful.

The result is GISMO (gastrointestinal smart module): an edible capsule about the size of a Tic Tac that travels the length of the human gut, takes a chemical reading every 20 seconds and sends the results to a receiver worn on the belt. It is an early example of a new generation of ingestible devices that can report back, live, from inside the gut. Potential applications run from routine diagnosis to targeted drug delivery, and eventually to electronics made entirely from food.

The GISMO measures changes in the gut's "redox balance", which can offer an early warning of inflamed or diseased tissue. Researchers are now testing it in people with ulcerative colitis and colorectal cancer. Patients swallow it before breakfast and must recover it, in the usual way, a few days later.

The human gut is an enormously complex ecosystem, home to trillions of microorganisms whose collective metabolic output offers a running commentary on their host's health. It has usually been difficult territory for doctors—endoscopy and colonoscopy can examine parts of it directly, but they are invasive. A standard colonoscopy takes around 30 minutes, costs several hundred

pounds, and is unpleasant enough that many patients who need one simply avoid it.

A swallowable camera called PillCam solved the visual part of that problem more than two decades ago. According to Medtronic, a firm based in Minneapolis which now owns the technology, PillCam has been used in more than four million patients worldwide. But the most important signals in the gut are often environmental: gases produced by microbes, acidity, the outputs of chemical reactions, inflammatory molecules, and how those conditions change with food, disease and drugs.

In 2018 a pilot study tested an ingestible capsule that was the first to send back readings of oxygen, hydrogen and carbon dioxide on its voyage through the hostile environment of the gut, whose acid is strong enough to dissolve metal and where conditions swing from hour to hour. The capsule could also detect changes in microbial fermentation after changes in the amount of dietary fibre. The goal since then has been to make such readings more precise and clinically useful.

To that end, researchers at the University of Maryland produced, in 2023, an ingestible capsule that

uses a gold electrode coated in Nafion, a polymer cousin of Teflon, to detect hydrogen sulphide in real time. The gas, responsible for the smell of rotten eggs, is produced both by gut bacteria involved in inflammatory bowel disease and, notably, by *Helicobacter pylori*, the bacterium behind most stomach ulcers and a major risk factor for gastric cancer.

The capsule was designed primarily to study gut inflammation, but the same approach could extend towards bacterial detection. One day it might replace current *H. pylori* diagnosis, which requires endoscopy, stool tests or sometimes imprecise breath analysis.

Ammonia can also be a flag for an *H. pylori* infection. In 2024 researchers at the University of Southern California developed an ingestible pill with optoelectronic sensors for oxygen and ammonia. Combined with neural-network algorithms it can map gas concentrations along the gastrointestinal tract with millimetre-scale precision. Animal trials are needed before human testing, but the goal is a capsule patients could use themselves, transmitting data to a smartphone.

Other gas-sensing capsules have already entered clinical development. Atmo Biosciences, an Aus-

tralian company, is running a trial of a capsule that measures fermentation gases to diagnose small intestinal bacterial overgrowth. The condition results from bacteria colonising the small intestine in abnormal numbers, fermenting nutrients before the body can absorb them. It causes bloating, pain and, in severe cases, malnutrition. According to the company, an earlier safety study found the capsule was 3,000 times more sensitive than standard breath testing.

Detection is only half the ambition. Others are aiming for edible electronics that can identify a disease signal, locate where in the gut it originates, and release treatment at that site. That would be better than dissolving drugs blindly in the stomach, which floods the whole body just to hit a specific patch of inflamed tissue.

Researchers at the Massachusetts Institute of Technology (MIT), for example, are developing ingestible capsules that sense internal conditions and act on what they find. In 2024 they created a device modelled on the jet propulsion of cephalopods, which pumps drugs directly into the wall of the digestive tract. Earlier that year, the MIT team had received \$66m from ARPA-H, a federal grant system that pushes high-risk, transfor-

mative health-care technology, to develop ingestible devices for the oral delivery of mRNA treatments. The five-year programme also seeks to develop electroceuticals, therapies based on electrical stimulation of the body's hormonal and neural signalling networks.

A challenge to all of this is power. Every sensor needs electricity and today's capsules rely on conventional batteries made from silver oxide—acceptable for a single diagnostic procedure, less so for routine monitoring of chronic conditions in millions of patients. A battery that enters the body must leave it as electronic waste. It is also often the capsule's largest component, setting a floor on miniaturisation.

The proposed solution is one that Willy Wonka would enjoy: electronic food. Researchers at the Italian Institute of Technology in Genoa have worked under a European Research Council project called ELFO (Electronic Food) to identify food-derived materials that can function as electronic components.

In 2023 they announced the world's first rechargeable edible battery, built from riboflavin (vitamin B2), quercetin (a flavonoid found in capers), activated charcoal, seaweed, beeswax and featuring

food-grade gold contacts. It operated at 0.65 volts and delivered 48 microamps for 12 minutes—plenty for a low-power LED, and enough to prove the principle.

The following year the same laboratory produced a fully edible transistor, built using copper phthalocyanine, a blue pigment found in toothpaste, as the semiconductor. Transistors are the fundamental switching elements of logic circuits. Edible versions are nowhere near as sophisticated as conventional chips, but that is arguably an easier problem to solve than asking people to eat bits of silicon.

Substantial obstacles remain. Edible batteries hold much less energy than lithium cells. Edible semiconductors are unstable and slow. Transmitting a wireless signal from inside the body is a persistent engineering issue because tissue absorbs radio waves. And a device that is simultaneously a food ingredient, a medical sensor, an electronic circuit and a drug delivery system is a bellyache for regulators.

It could, therefore, be a while before edible electronic devices reach pharmacies and supermarkets. But soon the gut could become a part of the

body that is able, when queried, to answer back.



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